Smart Pointers

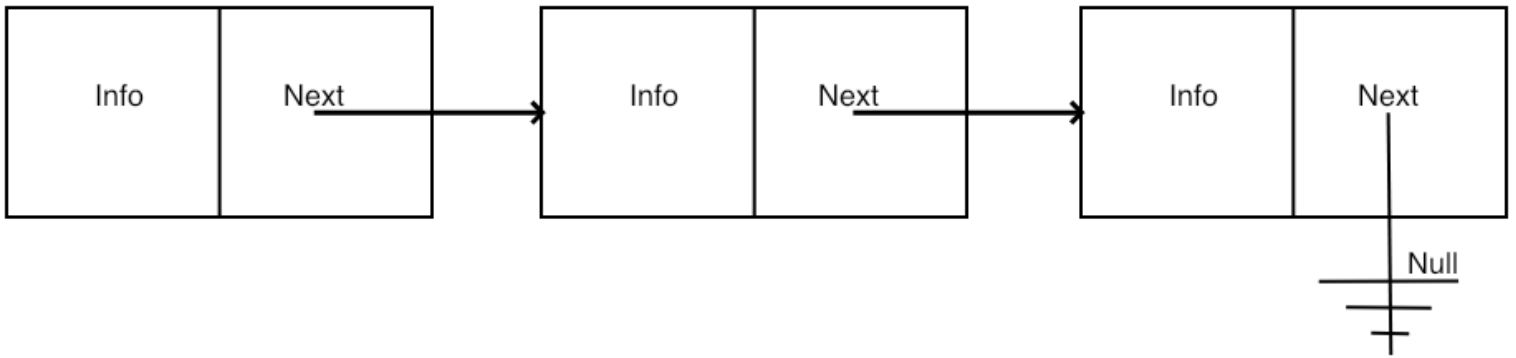
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Intro to pointers

1. Motivating application: linked list



- Used inside operating systems
- Model for complicated structures: trees, DAGs.

2. Recursive data structures

Naive code:

```
1 class Node {
2 private:
3   int value;
4   Node tail; // This does not work!
5   /* ... */
6 };
```

This does not work: would take infinite memory.

Indirect inclusion: only 'point' to the tail:

```
1 class Node {
2 private:
3   int value;
4   PointToNode tail;
5   /* ... */
6 };
```

3. Pointer types

- Smart pointers. You will see ‘shared pointers’.
- There are ‘unique pointers’. Those are tricky, but advanced users prefer them.
- Please don’t use old-style C pointers,
  but they have their uses.

Smart pointers

4. Example: step 3: headers to include

Using smart pointers requires at the top of your file:

```
1 #include <memory>
2 using std::shared_ptr;
3 using std::make_shared;
4
5 using std::unique_ptr;
6 using std::make_unique;
```

(unique pointers will not be discussed further here)

5. Example: step 1, we need a class

Simple class that stores one number:

Definition:

```
1 // pointer/pointx.cpp
2 class HasX {
3 private:
4   double x;
5 public:
6   HasX( double x) : x(x) {};
7   auto value() { return x; };
8   void set(double xx) {
9     x = xx; };
10 };
```

Example usage

```
1 // pointer/pointx.cpp
2 HasX xobj(5);
3 println("{}",xobj.value());
4 xobj.set(6);
5 println("{}",xobj.value());
```

6. Example: step 2, creating the pointer

Allocation of object and pointer to it in one:

```
1 auto X = make_shared<HasX>( /* constructor args */ );
2
3 // or explicitly:
4
5 shared_ptr<HasX> X =
6     make_shared<HasX>( /* constructor args */ );
```

7. Use of a shared pointer

Object vs pointed-object:

Code:

```
1 // pointer/pointx.cpp
2 #include <memory>
3 using std::make_shared;
4
5     /* ... */
6     HasX xobj(5);
7     println("{}",xobj.value());
8     xobj.set(6);
9     println("{}",xobj.value());
10
11    auto xptr = make_shared<HasX>(5);
12    println("{}",xptr->value());
13    xptr->set(6);
14    println("{}",xptr->value());
```

Output:

```
1 5
2 6
3 5
4 6
```

8. Example: step 4: in use

Set two pointers to the same object:

Code:

```
1 // pointer/twopoint.cpp
2 auto xptr = make_shared<HasX>(5);
3 auto yptr = xptr;
4 println( "{}",xptr->value() );
5 yptr->set(6);
6 println( "{}",xptr->value() );
```

Output:

```
1 5
2 6
```

What is the difference with

```
1 HasX xptr(5);
2 HasX yptr = xptr
3 cout << ...stuff...
```

?

9. Pointer dereferencing

Example: function

```
1 float distance_to_origin( Point p );
```

How do you apply that to a `shared_ptr<Point>`?
Use 'dereference' operator `*`:

```
1 shared_ptr<Point> p;
2 distance_to_origin( *p );
```

Exercise 1

Make a *DynRectangle* class, which is constructed from two shared-pointers-to-*Point* objects:

```
1 // pointer/dynrectangle.cpp
2 auto
3   origin  = make_shared<Point>(0,0),
4   fivetwo = make_shared<Point>(5,2);
5 DynRectangle lielow( origin,fivetwo );
```

Exercise 2

Test this design: Calculate the area, scale the top-right point, and recalculate the area:

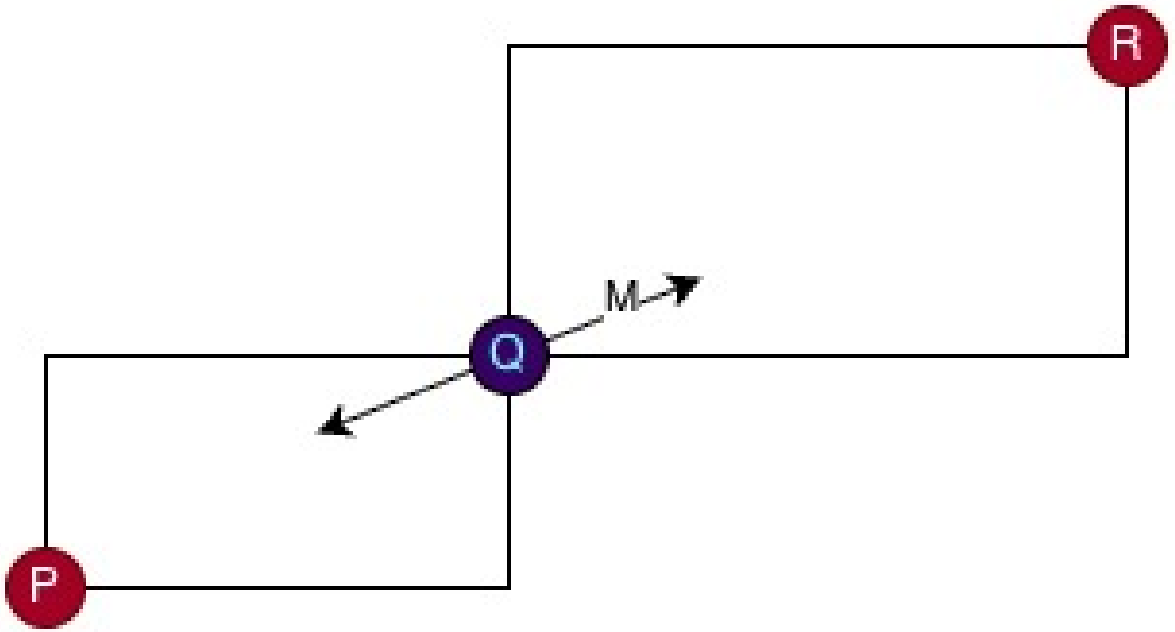
Code:

```
1 // pointer/dynrectangle.cpp
2 println( "Area: {}",lielow.area() );
3 /* ... */
4 println( "Area: {}",lielow.area() );
```

Output:

```
1 Area: 10
2 Area: 40
```

10. Illustration



Exercise 3

Make two *DynRectangle* objects so that the top-right corner of the first is the bottom-left corner of the other.

Now shift that point. Print out the two areas before and after to check correct behavior.

Use the *cxxopts* package to implement this syntax:

```
1 ./dynrectangle -p 1,1 -q 5,7 -r 10,12 -m 1,2
```

(referring to the previous picture)

Automatic memory management

11. Memory leaks

C has a 'memory leak' problem

```
1 // the variable `array' doesn't exist
2 {
3   // attach memory to `array':
4   double *array = new double[N];
5   // do something with array;
6   // forget to free
7 }
8 // the variable `array' does not exist anymore
9 // but the memory is still reserved.
```

The application 'is leaking memory'.
(even worse if you do this in a loop!)
Java/Python have 'garbage collection': runtime impact
C++ has the best solution: smart pointers with reference counting.

12. Illustration

We need a class with constructor and destructor tracing:

```
1 // pointer/ptr1.cpp
2 class thing {
3 public:
4   thing()  { cout << ".. calling constructor\n"; };
5   ~thing() { cout << ".. calling destructor\n"; };
6 };
```

13. Constructor / destructor in action

Code:

```
1 // pointer/ptr0.cpp
2 cout << "Outside\n";
3 {
4   thing x;
5   cout << "create done\n";
6 }
7 cout << "back outside\n";
```

Output:

```
1 Outside
2 .. calling
      ↪constructor
3 create done
4 .. calling destructor
5 back outside
```

14. Illustration 1: pointer overwrite

Let's create a pointer and overwrite it:

Code:

```
1 // pointer/ptr1.cpp
2 cout << "set pointer1"
3      << '\n';
4 auto thing_ptr1 =
5   make_shared<thing>();
6 cout << "overwrite pointer"
7      << '\n';
8 thing_ptr1 = nullptr;
```

Output:

```
1 set pointer1
2 .. calling
      ↪constructor
3 overwrite pointer
4 .. calling destructor
```

15. Illustration 2: pointer copy

Code:

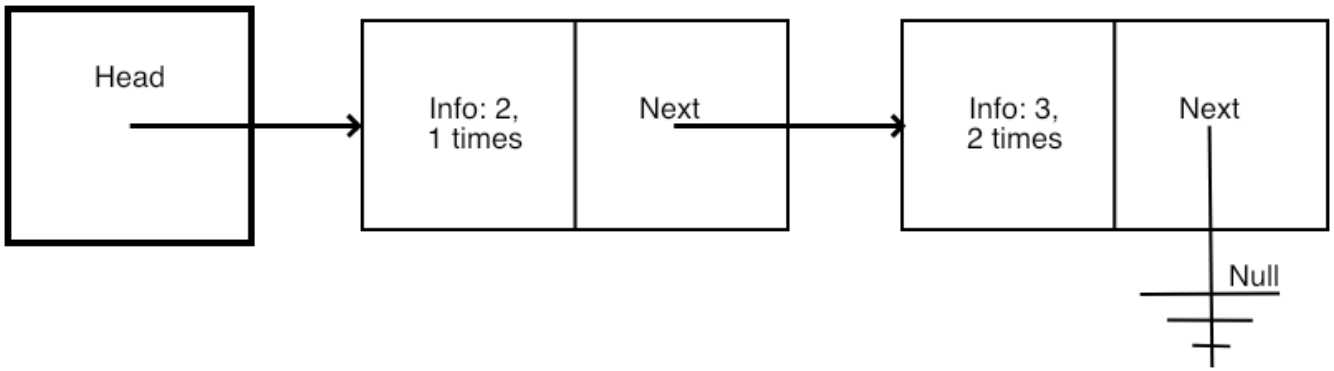
```
1 // pointer/ptr2.cpp
2 cout << "set pointer2" << '\n';
3 auto thing_ptr2 =
4   make_shared<thing>();
5 cout << "set pointer3 by copy"
6      << '\n';
7 auto thing_ptr3 = thing_ptr2;
8 cout << "overwrite pointer2"
9      << '\n';
10 thing_ptr2 = nullptr;
11 cout << "overwrite pointer3"
12      << '\n';
13 thing_ptr3 = nullptr;
```

Output:

```
1 set pointer2
2 .. calling
      ↪constructor
3 set pointer3 by copy
4 overwrite pointer2
5 overwrite pointer3
6 .. calling destructor
```

Example: linked lists

16. Linked list



17. Definition of List class

A linked list has as its only member a pointer to a node:

```
1 // tree/linkshared.cpp
2 class List {
3 private:
4   shared_ptr<Node> head{nullptr};
5 public:
6   List() {};
```

Initially null for empty list.

18. Definition of Node class

A node has information fields, and a link to another node:

```
1 // tree/linkshared.cpp
2 class Node {
3 private:
4   int datavalue{0},datacount{0};
5   shared_ptr<Node> next{nullptr};
6 public:
7   Node() {};
8   Node(int value,shared_ptr<Node> next=nullptr)
9     : datavalue(value),datacount(1),next(next) {};
```

A Null pointer indicates the tail of the list.

19. List methods

List testing and modification.

```
1  List mylist;
2  cout << "Empty list has length: "
3       << mylist.length() << '\n';
4
5  mylist.insert(3);
6  cout << "After one insertion the length is: "
7       << mylist.length() << '\n';
8  if (mylist.contains_value(3))
9    cout << "Indeed: contains 3" << '\n';
```

20. Print a list

Auxiliary function so that we can trace what we are doing.

Print the list head:

```
1 // tree/linkshared.cpp
2 void List::print() {
3   std::print("List:");
4   if (head!=nullptr)
5     std::print( " => "); head->
      print();
6   std::println("");
7 };
```

Print a node and its tail:

```
1 // tree/linkshared.cpp
2 void Node::print() {
3   std::print( "{}:{}",datavalue,
      datacount );
4   if (has_next()) {
5     std::print( ", " ); next->print
      ();
6   }
7 };
```

21. Recursive length computation

For the list:

```
1 // tree/linkshared.cpp
2 int List::length() {
3   if (head==nullptr)
4     return 0;
5   else
6     return head->length();
7 };
```

For a node:

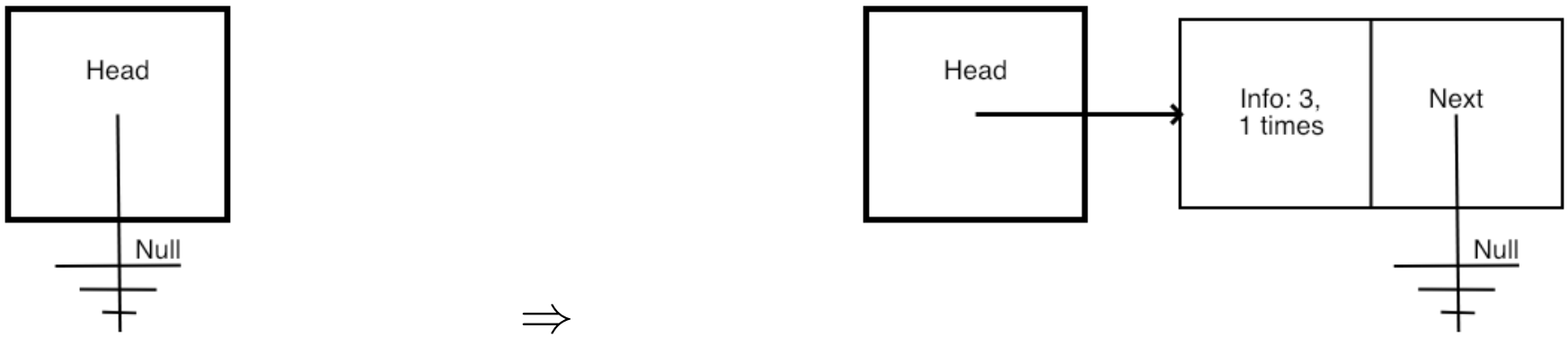
```
1 // tree/linkshared.cpp
2 int Node::length() {
3   if (!has_next())
4     return 1;
5   else
6     return 1+next->length();
7 };
```

Exercise 4

Test an empty list:

```
1 // catch/linktests.cpp
2 List mylist;
3 INFO( "Length of empty list" );
4 REQUIRE( mylist.length()==0 );
5 INFO( "Empty list has no headnode" );
6 REQUIRE_THROWS( mylist.headnode() );
```

22. Creating the first list element

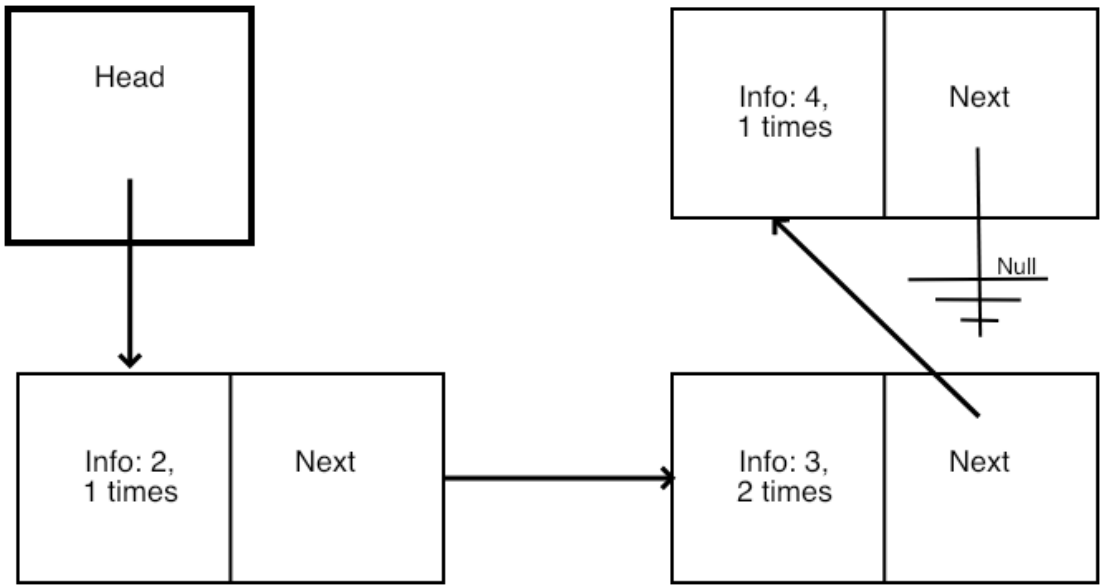


Exercise 5

Write the case of *Node*::**insert** that creates a node in an empty list.

```
1 // catch/linktests.cpp
2 mylist.insert(3);
3 REQUIRE( mylist.length()==1 );
4 REQUIRE( mylist.contains_value(3) );
5 REQUIRE( not mylist.contains_value(4) );
```

23. Element at the tail

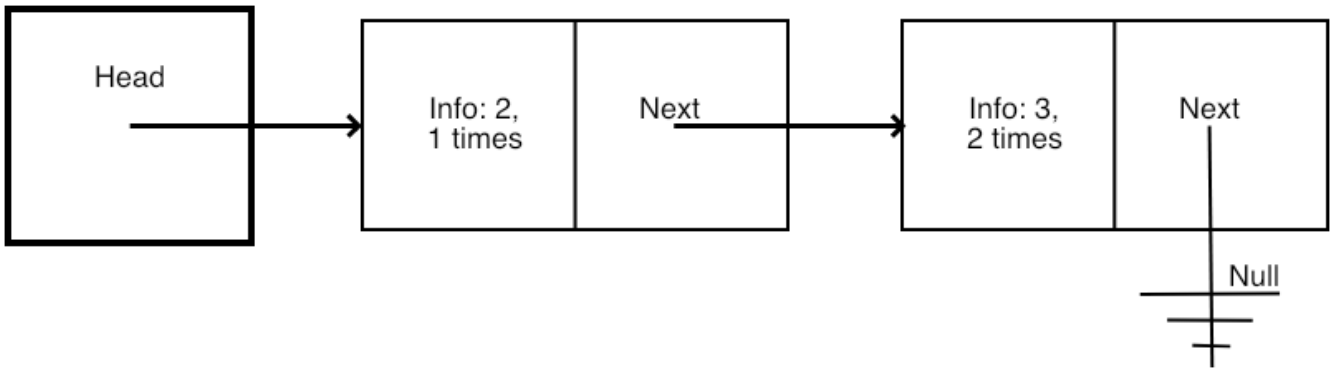


Exercise 6

If an item goes at the end of the list:

```
1 // catch/linktests.cpp
2 REQUIRE_NOTHROW( mylist.insert(6) );
3 REQUIRE( mylist.length()==3 );
4 REQUIRE( mylist.contains_value(6) );
```

24. Element at the head

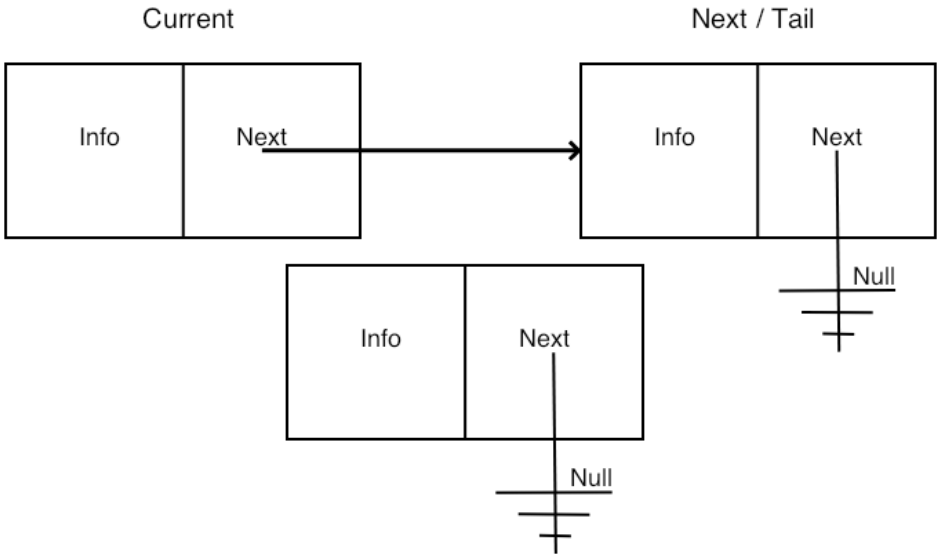


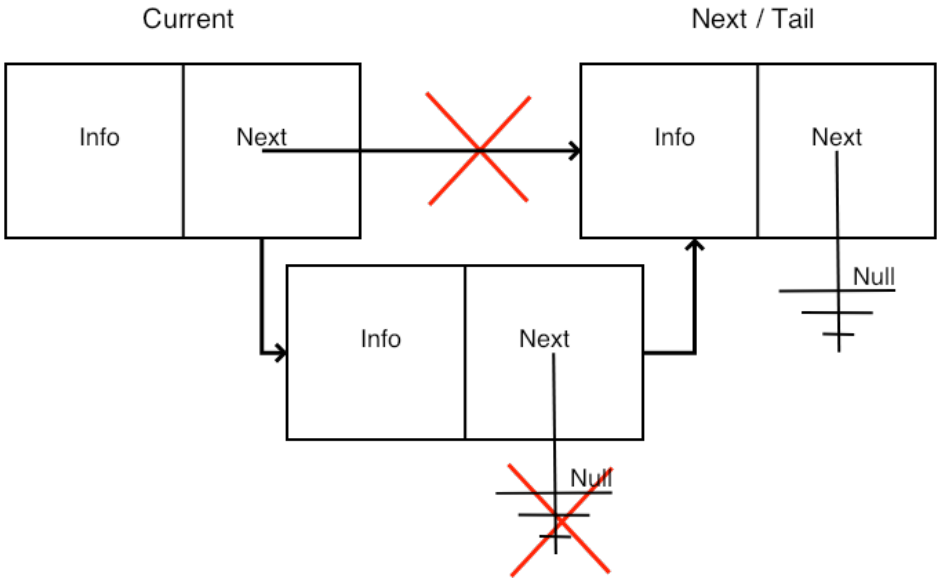
Exercise 7

One of the cases for inserting concerns an element that goes at the head. Update your `insert` method to get this to work:

```
1  // catch/linktests.cpp
2  REQUIRE_NOTHROW( mylist.insert(2) );
3  REQUIRE( mylist.length()==2 );
4  REQUIRE( mylist.contains_value(2) );
5  REQUIRE( mylist.contains_value(3) );
6
7  Node headnode;
8  INFO( "The head node should have 2, not 3" );
9  REQUIRE_NOTHROW( headnode = mylist.headnode() );
10 REQUIRE( headnode.value()==2 );
11 REQUIRE( headnode.count()==1 );
```

25. Insertion



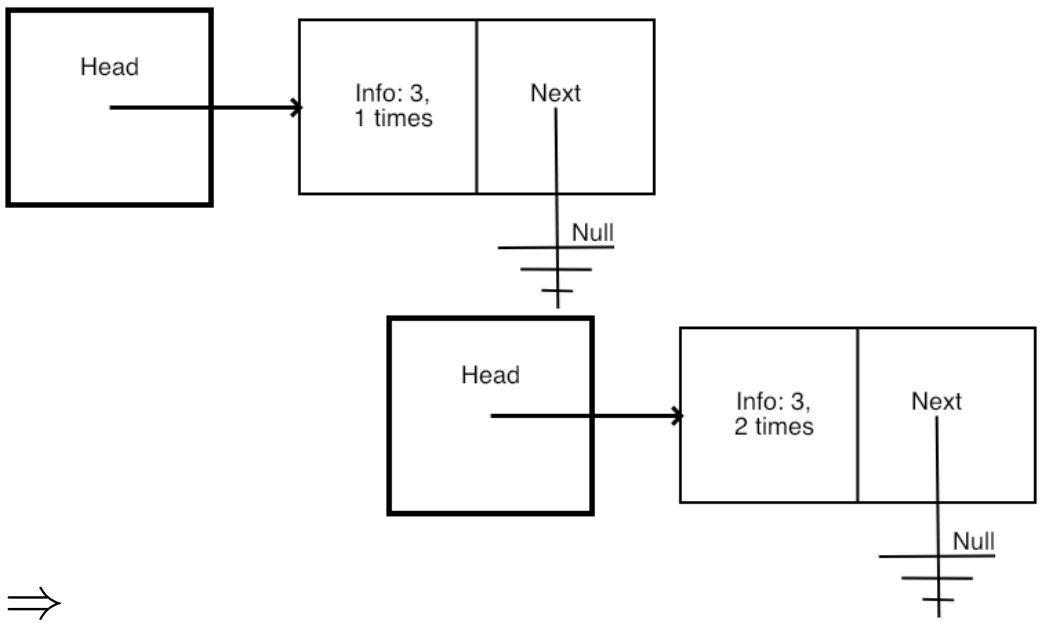


Exercise 8

Update your insert routine to deal with elements that need to go somewhere in the middle.

```
1 // catch/linktests.cpp
2 REQUIRE_NOTHROW( mylist.insert(4) );
3 REQUIRE( mylist.length()==4 );
4 REQUIRE( mylist.contains_value(4) );
5 REQUIRE( mylist.contains_value(3) );
```

26. Elements that are already in the list



$\Rightarrow$

Exercise 9

Inserting a value that is already in the list means that the `count` value of a node needs to be increased. Update your `insert` method to make this code work:

```
1 // catch/linktests.cpp
2 REQUIRE_NOTHROW( mylist.insert(3) );
3 REQUIRE( mylist.length()==1 );
4 REQUIRE( mylist.contains_value(3) );
5 Node headnode;
6 REQUIRE_NOTHROW( headnode = mylist.headnode() );
7 REQUIRE( headnode.value()==3 );
8 REQUIRE( headnode.count()==2 );
```

Exercise 10

Write a program that accepts positive integers from interactive input;

- The program builds a linked list of its inputs, sorted by increasing size.
- If an input value occurs multiple times, it should be counted.
- The program should stop if the input is zero; then it prints the resulting list.

Example: inputs `5,7,5,8,0` should give

```
5:2 7:1 8:1
```

Iterative algorithms

27. Iterative computation of the list length

Use a shared pointer to go down the list:

```
1  // tree/linkshared.cpp
2  int List::length_iterative() {
3    int count = 0;
4    if (head!=nullptr) {
5      auto current_node = head;
6      while (current_node->has_next()) {
7        current_node = current_node->nextnode(); count += 1;
8      }
9    }
10   return count;
11 };
```

(Fun exercise: can do an iterative de-allocate of the list?)

Unique pointers